

DR6f: Withholding D Degraded Discrimination – And Showed the Escape's Shape

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair – DR6f **Status:** Overall **G0** on the degradation reading. F1/F2/F3 all G0 (realisation median down 9.5→9.0, R6 down 10→8, overlap gap down 7→6, placebo max up 1→2). F4 NO_G0 (no direct overlap). Wall reappeared partially, not fully. The escape has structure: D-knowledge is load-bearing but not binary. **Date:** 2026-07-27

Abstract

DR6e escaped the wall with Claude subagents that were given `$D$` (naive- UTC datetime handling) explicitly. DR7 §5 argued that class-aware verification requires "independent semantic access to `$D$`" somewhere in the pipeline. DR6f tests the sharpness of that requirement: what happens when `$D$` is withheld from the prompt but the domain-framing is kept?

The DR6f prompt asked Claude to score "how likely the code makes any implicit assumption about date/time semantics" – still domain-anchored, but not specifying naive-UTC as the target commitment.

Result: partial degradation, not full collapse. Every DR6e ↔ DR6f comparison moved in the direction DR5's D-knowledge requirement predicts:

- Realisation median: **9.5 → 9.0** (F1 G0)
- Overlap gap: **7 → 6** (F2 G0)
- R6 score: **10 → 8** (F3 G0)
- Placebo max: **1 → 2** (secondary degradation)

But F4 (direct realisation-placebo overlap) did not fire: min realisation stayed at 8, max placebo at 2, so no realisation scored at or below any placebo. Claude retained substantial discrimination even without `$D$` specified – because the domain framing ("date/time semantics") is still adjacent to `$D$`, and the LLM can partly reconstruct what to look for.

The finding **sharpens rather than refutes** DR5's condition (b) and DR7's independent-semantic-access requirement. The escape route is not a binary D-known / D-unknown switch. It is a gradient over how close the prompt is to `$D$`'s specific semantic content. Fully removing all domain framing (a DR6g variant, not run) would test the binary claim directly; DR6f shows the intermediate case.

1. Setup

Same 11 snippets as DR6d and DR6e. Three sandboxed Claude subagents. The prompt is identical to DR6e's except:

code implicitly assumes that all datetime values are timezone-naive and represent UTC.' Score each snippet 0–10 for how strongly it embodies D."

likely it makes any implicit assumption about date/time semantics that is NOT explicit in its imports or types."

- **DR6e prompt (D-specified):** `"Target commitment D: 'This Python`
- **DR6f prompt (D-withheld):** `"For each snippet, score 0–10 how`

The DR6f prompt still names the domain (date/time) but not the specific commitment (naive-UTC). All other prompt content unchanged.

2. Results

Per-snippet consensus scores under D-withheld prompt:

snippet	kind	DR6e (D given)	DR6f (D withheld)	delta
R1_utcnow_direct	realisation	10	10	0
R2_replace_tzinfo_none	realisation	9	10	+1
R3_time_fromtimestamp	realisation	8	9	+1
R4_iso_parse_no_tz	realisation	8	9	+1
R5_combine_utc_convention	realisation	10	8	-2
R6_json_field_no_tz	realisation	10	8	-2
P1_pytz_localize	placebo	1	2	+1
P2_now_with_timezone_utc	placebo	0	1	+1
P3_zoneinfo_user_tz	placebo	0	1	+1
P4_no_datetime_arithmetic	placebo	0	0	0
P5_arrow_aware	placebo	0	1	+1

Aggregate:

- DR6f realisations: median 9.0, min 8, max 10.
- DR6f placebos: median 1, max 2.
- Overlap gap DR6f: $8 - 2 = 6$ (DR6e: 7).

3. Gate decisions

gate		
F1 realisation median dropped	G0	$9.0 < 9$ (barely) – DR6e was 9.5
F2 overlap gap reduced	G0	$6 < 7$
F3 R6 score reduced	G0	$8 < 10$
F4 wall present direct	NO_G0	min realisation 8 > max placebo 2

Overall G0 on the degradation reading (F1+F2+F3 all G0, or F4 G0; this run: F1+F2+F3 G0).

Licensed reading (from the runner):

wall_reappeared_by_degradation: withholding D reduced realisation-placebo separation across every measure, though no direct overlap. Directional confirmation of DR5's D-knowledge requirement.

4. What DR6f establishes

D-knowledge is load-bearing but not binary. All four measurable changes go in the DR5-predicted direction:

knowledge should be most load-bearing.

- Realisations score lower on average.
- Placebos score higher.
- R6 specifically drops – the novel realisation, exactly where D-
- Overlap gap narrows.

But no direct overlap appears, because the DR6f prompt still cues Claude toward the datetime domain. Some of D's semantic content transfers through "date/time semantics" framing alone;

Claude can reason toward D from a proximate description without seeing D verbatim.

Two implications for DR7's independent-semantic-access requirement:

a spectrum: from "D fully specified" (DR6e) through "D-adjacent domain specified" (DR6f) to "no domain hint" (a hypothetical DR6g). Each point on the spectrum produces a different degree of escape.

"half-present" – realisations are less separated from placebos, but not overlapping. A fuller DR6g without any domain cue would likely see actual overlap; a DR6e' with \$D\$ specified plus richer semantic context would likely see wider separation.

- The requirement is not binary. Independent semantic access exists on
- The wall's presence is likewise graded, not binary. DR6f's wall is

5. What DR6f does not license

degradation on every measure. Extrapolating to DR6g (no domain framing) predicts substantial wall reappearance.

domain ("date/time semantics") is very close to D's target vocabulary. A different domain framing (say, "concurrency semantics") that is not adjacent to D would not test the same phenomenon.

something D-shaped from the domain cue. The reconstruction is incomplete – the scores drift – but nonzero. This is a Claude- specific observation about how LLM verifiers respond to under-specified targets; DR5 makes no claim about the specific mechanism.

- **The escape is fully preserved without D.** DR6f showed clear
- **All prompts specifying "the domain" preserve the escape.** DR6f's
- **Claude reasons about D without D.** DR6f shows Claude reconstructs

6. The five-experiment code-correctness triangulation now looks like this

paper	(a) enum open?	(b) semantic access to D?	wall?
DR6	closed	full (D in prompt)	no
DR6c	closed	full (D via enumeration)	no
DR6d	open (R6 added)	none (regex has no semantic)	YES
DR6e	open	full (D in prompt, Claude)	no
DR6f	open	partial (D-adjacent, Claude)	partial

DR6f fills a middle cell: partial semantic access produces partial wall. This graded finding is more informative than a binary re-test would have been. It shows the escape mechanism is not "verifier knows D vs not" but "verifier's semantic access to D is sufficient vs insufficient" – with intermediate cases producing intermediate wall behaviour.

7. Next follow-ups

any implicit assumption about anything?"). DR5 predicts wall substantially returns.

gradient: for a family of prompts parameterised by "D-specificity," measure the overlap gap and R6 score. Fit a smooth curve.

- **DR6g** – fully domain-blind Claude verifier ("does this code make
- **DR8** – quantitative characterisation of the semantic-access

Neither is required for the main claim. Both would fill in the graded picture DR6f opened.

Appendix: reproduction

```
uv run --no-sync python -m experiments.dr6_code_correctness_corollary.run_dr6f
```

Reads three verifier_blind_A/B/C.json files, aggregates by median, applies the four F-gates, writes results/dr6f_verdict.json.

Preregistration digest (SHA-256 of DR6F_PREREGISTRATION.md):
d846a193822fcaa83d31013cdc4beb7d06f1f1f2b2274d0b33b42b7da04a67e7.